

Module - 5

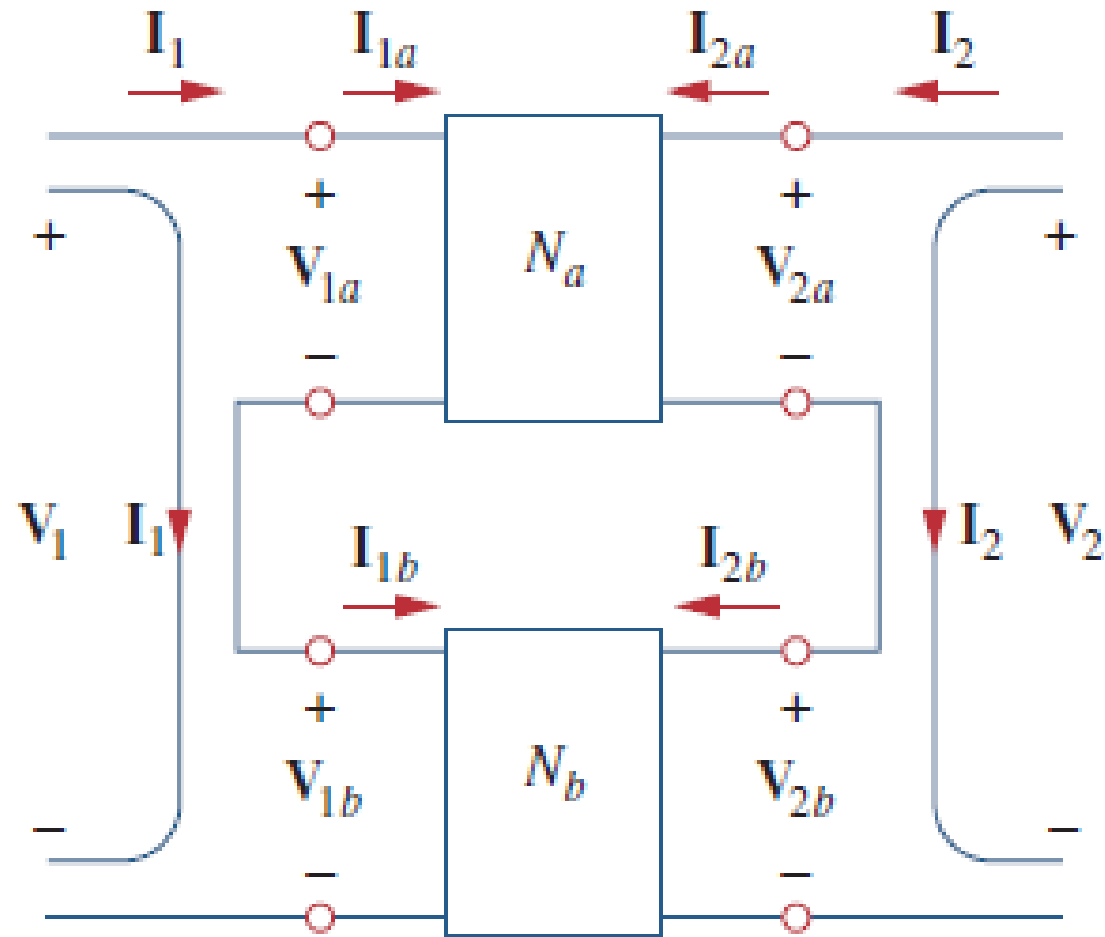
Interconnection of Networks

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Interconnection of Networks

- A large complex network may be divided into subnetworks for the purpose of analysis and design.
- The subnetworks are modelled as two-port networks, interconnected to form the original network.
- The two port networks may therefore be regarded as building blocks that can be interconnected to form a complex network.
- The interconnection can be in series, in parallel, or in cascade.
- Series networks – Z parameters
- Parallel networks – Y parameters
- Cascade networks – Transmission parameters

Series connection of two two-port networks



Series connection of two two-port networks

- The networks are regarded as being in series because their input currents are the same and their voltages add.
- Each network has a common reference, and when the circuits are placed in series, the common reference points of each circuit are connected together.

- For Network N_a ,

$$V_{1a} = z_{11a}I_{1a} + z_{12a}I_{2a}$$

$$V_{2a} = z_{21a}I_{1a} + z_{22a}I_{2a}$$

- For Network N_b ,

$$V_{1b} = z_{11b}I_{1b} + z_{12b}I_{2b}$$

$$V_{2b} = z_{21b}I_{1b} + z_{22b}I_{2b}$$

- From the figure,

$$\mathbf{I}_1 = \mathbf{I}_{1a} = \mathbf{I}_{1b}, \quad \mathbf{I}_2 = \mathbf{I}_{2a} = \mathbf{I}_{2b}$$

$$\mathbf{V}_1 = \mathbf{V}_{1a} + \mathbf{V}_{1b} = (\mathbf{z}_{11a} + \mathbf{z}_{11b})\mathbf{I}_1 + (\mathbf{z}_{12a} + \mathbf{z}_{12b})\mathbf{I}_2$$

$$\mathbf{V}_2 = \mathbf{V}_{2a} + \mathbf{V}_{2b} = (\mathbf{z}_{21a} + \mathbf{z}_{21b})\mathbf{I}_1 + (\mathbf{z}_{22a} + \mathbf{z}_{22b})\mathbf{I}_2$$

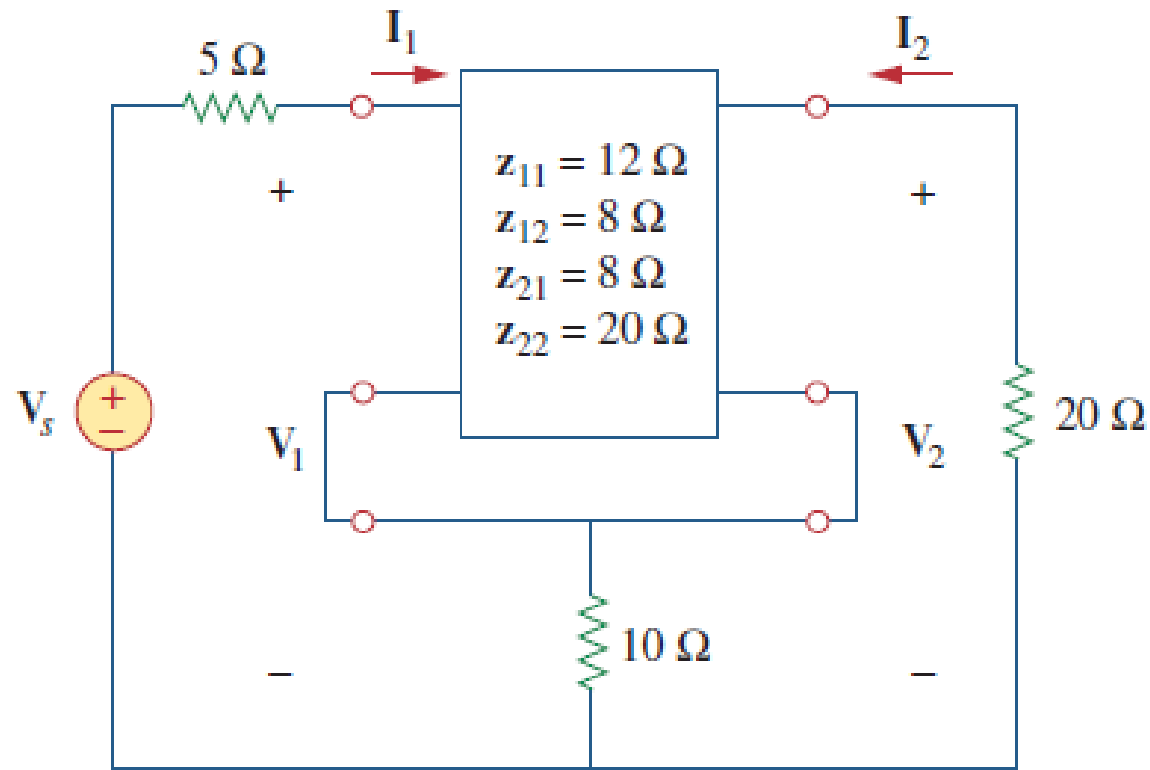
- Thus the Z parameter for the overall network is

$$\begin{bmatrix} \mathbf{z}_{11} & \mathbf{z}_{12} \\ \mathbf{z}_{21} & \mathbf{z}_{22} \end{bmatrix} = \begin{bmatrix} \mathbf{z}_{11a} + \mathbf{z}_{11b} & \mathbf{z}_{12a} + \mathbf{z}_{12b} \\ \mathbf{z}_{21a} + \mathbf{z}_{21b} & \mathbf{z}_{22a} + \mathbf{z}_{22b} \end{bmatrix}$$

$$[\mathbf{z}] = [\mathbf{z}_a] + [\mathbf{z}_b]$$

Numerical - 1

Evaluate V_2/V_s in the circuit



Solution:

This may be regarded as two two-ports in series. For N_b ,

$$z_{12b} = z_{21b} = 10 = z_{11b} = z_{22b}$$

Thus,

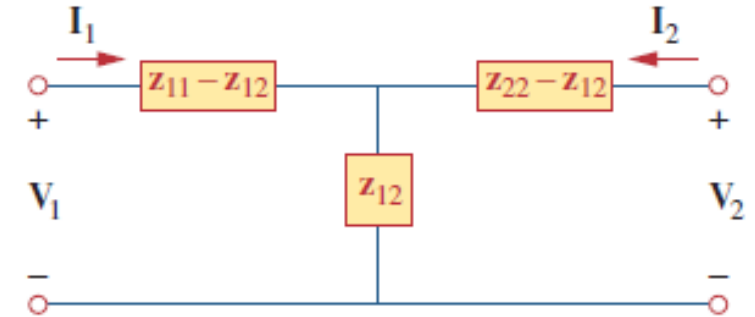
$$[z] = [z_a] + [z_b] = \begin{bmatrix} 12 & 8 \\ 8 & 20 \end{bmatrix} + \begin{bmatrix} 10 & 10 \\ 10 & 10 \end{bmatrix} = \begin{bmatrix} 22 & 18 \\ 18 & 30 \end{bmatrix}$$

$$V_1 = z_{11}I_1 + z_{12}I_2 = 22I_1 + 18I_2$$

$$V_2 = z_{21}I_1 + z_{22}I_2 = 18I_1 + 30I_2$$

Also, at the input port

$$V_1 = V_s - 5I_1$$



and at the output port

$$\mathbf{V}_2 = -20\mathbf{I}_2 \quad \Rightarrow \quad \mathbf{I}_2 = -\frac{\mathbf{V}_2}{20}$$

$$\mathbf{V}_s - 5\mathbf{I}_1 = 22\mathbf{I}_1 - \frac{18}{20}\mathbf{V}_2 \quad \Rightarrow \quad \mathbf{V}_s = 27\mathbf{I}_1 - 0.9\mathbf{V}_2$$

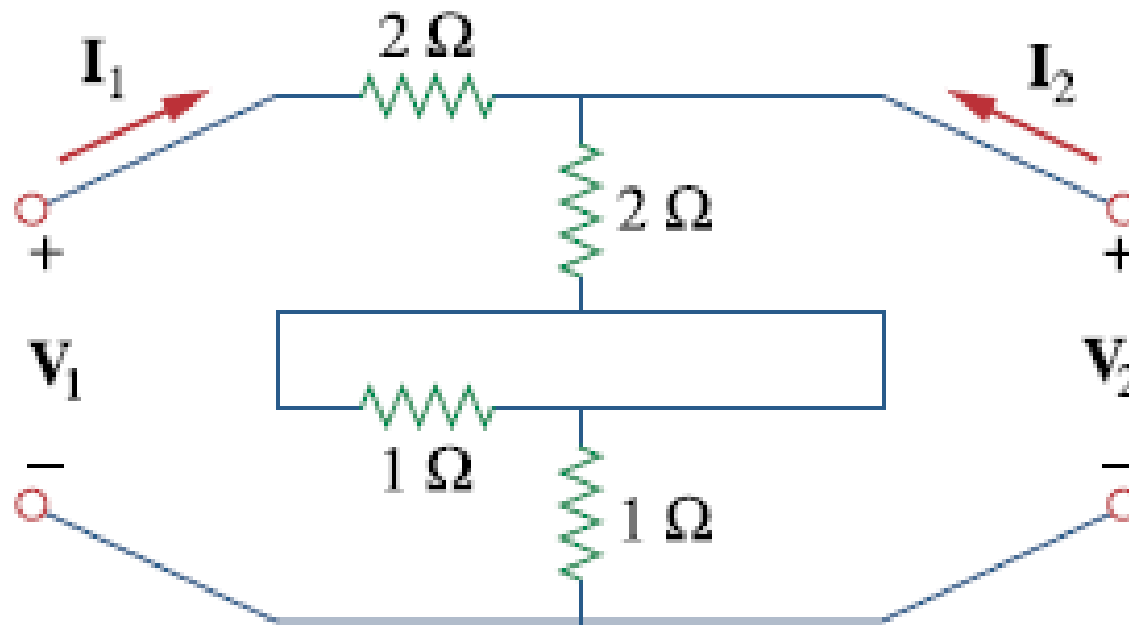
$$\mathbf{V}_2 = 18\mathbf{I}_1 - \frac{30}{20}\mathbf{V}_2 \quad \Rightarrow \quad \mathbf{I}_1 = \frac{2.5}{18}\mathbf{V}_2$$

$$\mathbf{V}_s = 27 \times \frac{2.5}{18}\mathbf{V}_2 - 0.9\mathbf{V}_2 = 2.85\mathbf{V}_2$$

$$\frac{\mathbf{V}_2}{\mathbf{V}_s} = \frac{1}{2.85} = 0.3509$$

Numerical - 2

What is the y parameter presentation of the circuit



Solution The network consists of two two-ports in series. It is better to work with z parameters and then convert to y parameters.

$$\text{For } N_a, \quad [z_a] = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$$

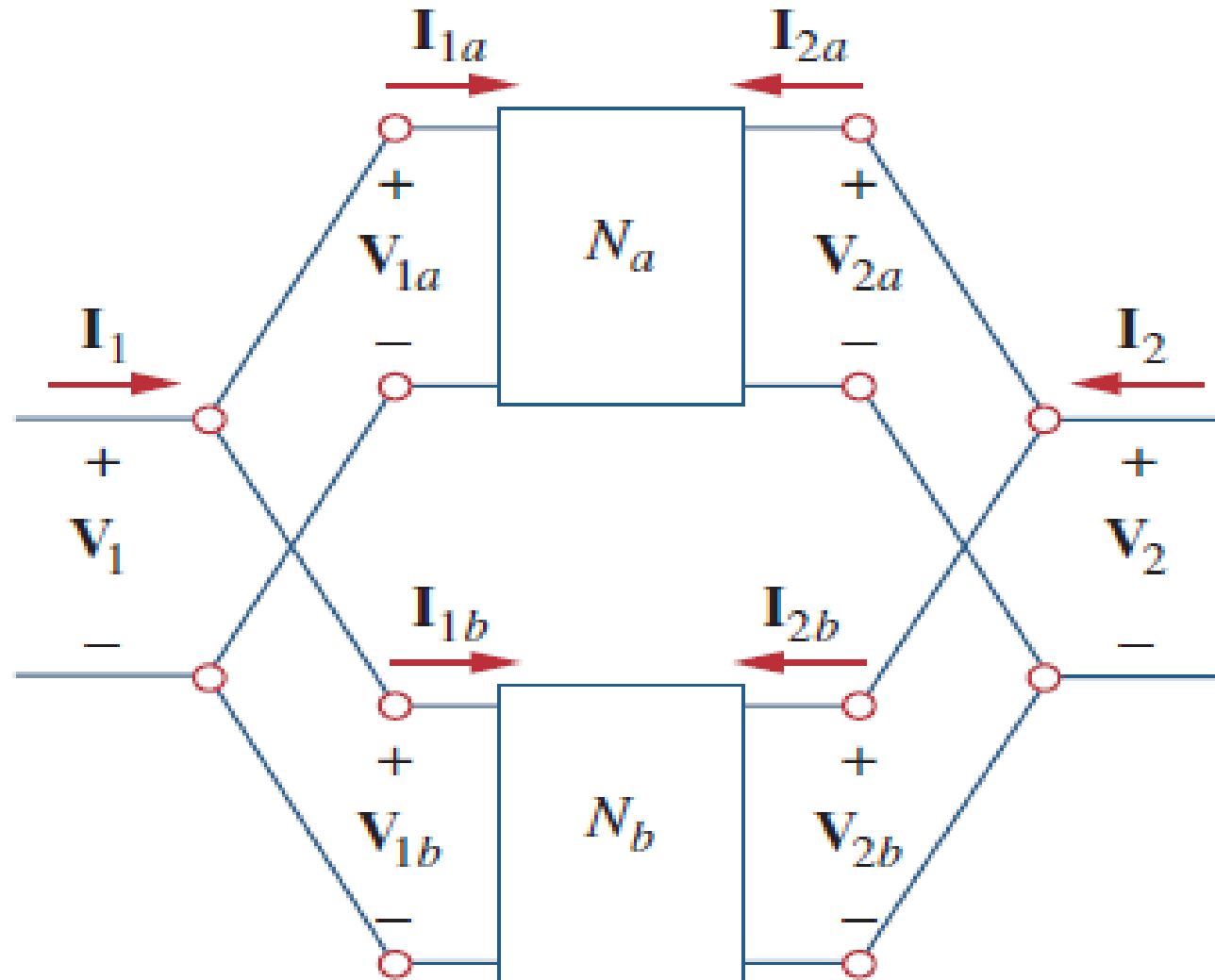
$$\text{For } N_b, \quad [z_b] = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$[z] = [z_a] + [z_b] = \begin{bmatrix} 6 & 3 \\ 3 & 3 \end{bmatrix}$$

$$\Delta_z = 18 - 9 = 9$$

$$[y] = \begin{bmatrix} \frac{z_{22}}{\Delta_z} & \frac{-z_{12}}{\Delta_z} \\ \frac{-z_{21}}{\Delta_z} & \frac{z_{11}}{\Delta_z} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{-1}{3} \\ \frac{-1}{3} & \frac{2}{3} \end{bmatrix} \text{ S}$$

Parallel connection of two two-port networks



Parallel connection of two two-port networks

- Two two-port networks are in parallel when their port voltages are equal and the port currents of the larger network are the sum of the individual port currents.
- Each circuit must have a common reference and when the networks are connected together they must all have their common references tied together.

- For Network N_a ,

$$I_{1a} = y_{11a}V_{1a} + y_{12a}V_{2a}$$

$$I_{2a} = y_{21a}V_{1a} + y_{22a}V_{2a}$$

- For Network N_b ,

$$I_{1b} = y_{11b}V_{1b} + y_{12b}V_{2b}$$

$$I_{2b} = y_{21b}V_{1b} + y_{22b}V_{2b}$$

$$V_1 = V_{1a} = V_{1b}, \quad V_2 = V_{2a} = V_{2b}$$

$$I_1 = I_{1a} + I_{1b}, \quad I_2 = I_{2a} + I_{2b}$$

$$I_1 = (y_{11a} + y_{11b})V_1 + (y_{12a} + y_{12b})V_2$$

$$I_2 = (y_{21a} + y_{21b})V_1 + (y_{22a} + y_{22b})V_2$$

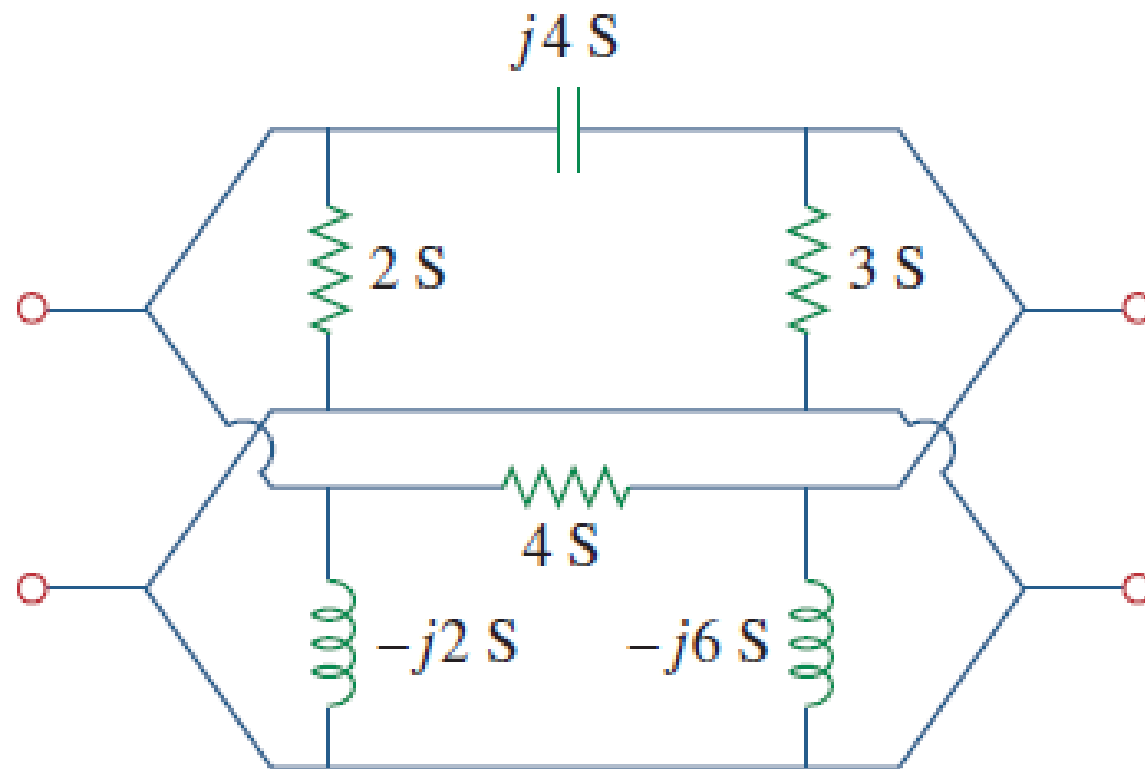
Thus, the y parameters for the overall network are

$$\begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \begin{bmatrix} y_{11a} + y_{11b} & y_{12a} + y_{12b} \\ y_{21a} + y_{21b} & y_{22a} + y_{22b} \end{bmatrix}$$

$$[y] = [y_a] + [y_b]$$

Numerical - 1

Find the y parameters of the two-port



Solution:

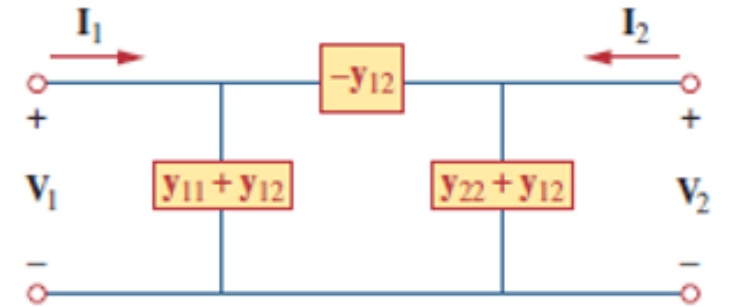
Let us refer to the upper network as N_a and the lower one as N_b .

$$y_{12a} = -j4 = y_{21a}, \quad y_{11a} = 2 + j4, \quad y_{22a} = 3 + j4$$

$$[y_a] = \begin{bmatrix} 2 + j4 & -j4 \\ -j4 & 3 + j4 \end{bmatrix}$$

$$y_{12b} = -4 = y_{21b}, \quad y_{11b} = 4 - j2, \quad y_{22b} = 4 - j6$$

$$[y_b] = \begin{bmatrix} 4 - j2 & -4 \\ -4 & 4 - j6 \end{bmatrix}$$

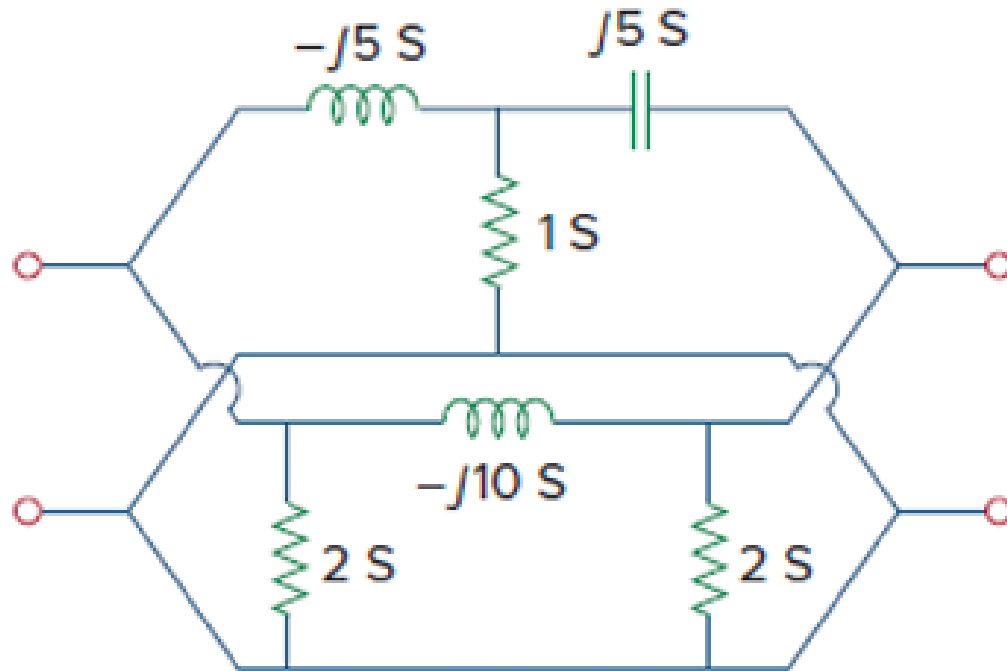


The overall y parameters are

$$[y] = [y_a] + [y_b] = \begin{bmatrix} 6 + j2 & -4 - j4 \\ -4 - j4 & 7 - j2 \end{bmatrix}$$

Numerical - 2

Find the y parameters of the two-port



Answer:

$$[y] = \begin{bmatrix} 27 - j15 & -25 + j10 \\ -25 + j10 & 27 - j5 \end{bmatrix} \text{ S}$$

Numerical - 3

Obtain the h parameters for the network

